Overcoming High-Frequency Attenuation in UAV Infrared–Visible Remote Sensing Tiny-Object Detection: Frequency-Domain Amplitude–Phase Decoupling Network

Xiaomin Liu, *Member, IEEE*, Haojie Yin, and Haoyu Wang, *Member, IEEE*

***Abstract*—Tiny objects in multimodal uncrewed aerial vehicle (UAV) remote sensing images are highly susceptible to being overwhelmed by the high-frequency clutter of complex terrains. Existing visible–infrared fusion methods are predominantly confined to the spatial domain, where direct interaction between large-area infrared thermal responses and fine visible structures may attenuate the structural cues required for tiny-object detection. To address structural degradation during multimodal feature interaction, this article proposes a frequency-domain amplitude–phase decoupling network (FADN) for UAV-based Red Green Blue–Infrared (RGB-IR) tiny-object detection. FADN introduces frequency-domain interaction as a complementary representation to spatial-domain feature learning. In particular, an uncertainty-aware phase–amplitude cross-modulation module employs the visible phase as the dominant structural anchor and incorporates complementary infrared amplitude information together with a bounded phase residual according to the estimated infrared confidence. A radial frequency-band recalibration mechanism is further introduced to adaptively model the responses of different radial frequency bands. In addition, a frequency–spatial synergistic interaction module is constructed, in which frequency-domain responses guide adaptive spatial feature sampling, while spatial contextual information modulates the frequency-domain representation. Finally, a frequency-domain structural constraint loss function is designed. By combining RGB-phase consistency, infrared-confidence regularization, and a cross-domain diversity term based on absolute cosine similarity, it regulates multimodal feature learning and reduces representation redundancy. Extensive experiments on the large-scale cross-modal remote sensing datasets DroneVehicle and Visible-Thermal Uncrewed Aerial Vehicle (VTUAV)-det demonstrate that FADN effectively alleviates the high-frequency feature decay problem and significantly improves the detection performance of tiny objects in complex remote sensing scenarios.**

***Index Terms*—Frequency domain, multimodal object detection, remote sensing, tiny object, uncrewed aerial vehicle (UAV).**

Received 2 June 2026; revised 1 September 2026; accepted 2 October 2026. Date of publication 13 October 2026; date of current version 2 November 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62303468 and Grant 62303469, in part by the Science and Technology Program of Xuzhou under Grant KC25123, and in part by China University of Mining and Technology Trained the Eleventh Group of University-Level Outstanding Young Backbone Teachers. *(Corresponding author: Haoyu Wang.)*

The authors are with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: xiaominliu@cumt.edu.cn; yinhaojiecumt@163.com; wanghaoyucumt@163.com).

Digital Object Identifier 10.1109/TGRS.2026.3742556

I. Introduction

In recent years, with the rapid development of uncrewed aerial vehicle (UAV) technology, remote sensing object detection from a UAV perspective has demonstrated significant application value in fields such as intelligent inspection [1], [2], [3], disaster rescue [4], [5], and security monitoring [6], [7]. However, in complex aerial remote sensing scenarios, tiny objects typically occupy only a few dozen pixels and are easily overwhelmed by high-frequency background clutter, such as topographical textures and water surface reflections [8], [9], [10]. Although Red Green Blue (RGB) sensors can provide high-resolution texture and structural details, they tend to lose contrast and details under adverse illumination changes, such as intense midday sunlight or dim dusk light [11], [12], [13]. To this end, multimodal perception systems commonly introduce Infrared (IR) images to capture complementary thermal radiation information, thereby compensating for the perceptual deficiencies of visible light in degraded environments [14], [15]. Constructing an RGB–IR multimodal fusion perception system is an effective approach to alleviate the challenges of all-weather remote sensing tiny-object detection.

Despite their complementary characteristics, RGB and infrared modalities remain difficult to exploit effectively for UAV-based tiny-object detection. Tiny target responses are easily weakened by repeated downsampling and overwhelmed by complex backgrounds, while variations in flight altitude further aggravate scale differences. In addition, visible images are sensitive to illumination and weather conditions, whereas infrared images often suffer from thermal crossover, weak boundaries, and limited texture. These modality-specific degradations, together with residual cross-modal misalignment, make effective multimodal feature interaction particularly challenging.

Although existing multimodal feature fusion methods have made certain progress, most of them rely on feature alignment and interaction within the spatial domain, such as pixel-level weighting or spatial attention-based concatenation [16], [17].

Related studies further cover spectral-bias analysis and suppression [18], [19], illumination-guided RGB–IR alignment [20], the influence of high-frequency components on neural-network learning [21], [22], and frequency–spatial modeling

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

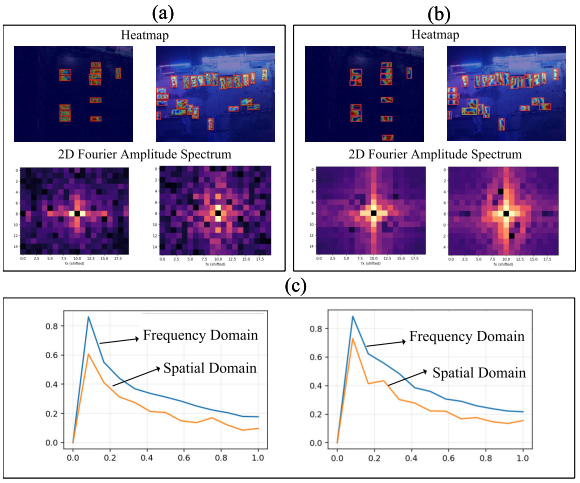


Fig. 1. Necessity of multiview analysis and frequency-domain reconstruction of spatial ambiguity and high-frequency attenuation feature degradation in existing RGB–IR fusion methods. (a) Spatial domain. (b) Frequency domain. (c) Radial mean spectrum curve.

for infrared target detection and remote sensing change detection [23], [24].

These spatial-domain operations directly mix structural cues and modality-dependent energy responses within a common feature representation. In UAV scenes, object contours, edges, and fine details are closely associated with phase organization and mid-to-high-frequency variations, whereas large-area infrared thermal responses may contain substantial low-frequency energy. Direct interaction between these heterogeneous responses may weaken fine visible structures and reduce the discriminative capability of the fused representation for tiny-object detection in cluttered backgrounds.

Existing frequency-aware detection methods have demonstrated the effectiveness of frequency enhancement, high- and low-frequency decomposition, and frequency-band reweighting. However, most of these methods treat frequency information as a general enhancement cue and remain limited in explicitly assigning asymmetric roles to visible and infrared amplitude–phase components. The contribution of unreliable infrared responses is also rarely regulated according to modality confidence. Therefore, the central research gap addressed in this work is how to establish a modality-asymmetric and confidence-controlled frequency-domain interaction mechanism that maintains visible-dominant structural cues while selectively incorporating reliable infrared information.

To intuitively verify the effectiveness of frequency-domain representation in suppressing backgrounds and preserving high-frequency details, a comparative analysis is conducted from the perspectives of spatial feature responses and frequency-domain energy distributions, as shown in Fig. 1. The feature heatmaps generated by spatial fusion-based methods exhibit diffusion and blurring phenomena, whereas features based on frequency-domain modulation demonstrate clear boundary localization and significant background suppression, which implies that the edge sharpness of the targets is effectively preserved. From the frequency-domain perspective, a semiquantitative analysis of the 2-D Fourier amplitude spectrum reveals that this spatial blurring essentially originates from the dispersion and systematic decay of noncentral region energy during the fusion process. Further quantitative verification using radial mean spectrum curves demonstrates that in the critical mid-to-high frequency bands, the feature energy of spatial methods drops sharply as frequency increases, decaying almost to zero in the high-frequency band, which is exactly the crucial reason leading to missed and false detections of tiny objects from a UAV perspective. On the contrary, the proposed frequency-domain amplitude–phase decoupling network (FADN) can effectively preserve abundant high-frequency discriminative information. The above analysis clearly indicates that high-frequency decay is the core bottleneck restricting the performance of existing methods, establishing a solid theoretical foundation for the network design in this article.

Based on the above insights, this article proposes FADN for multimodal remote sensing tiny-object detection from a UAV perspective.

Rather than replacing spatial-domain feature learning, FADN introduces complementary frequency-domain interaction into the multimodal fusion process. In particular, the visible phase serves as the dominant structural anchor, while infrared amplitude information and a bounded phase residual are incorporated under the control of infrared confidence. On this basis, radial frequency-band responses are adaptively recalibrated, and bidirectional interaction is established between frequency-domain responses and spatial features.

The entire framework is optimized end-to-end by a frequency-domain structural constraint loss function. The main contributions of this article are summarized as follows.

1) FADN is proposed to alleviate structural degradation during approximately aligned RGB–IR feature fusion. Its UACM module retains the visible phase as the dominant structural anchor while incorporating confidence-controlled infrared amplitude information and a bounded phase residual.
2) RFBR independently recalibrates radial frequency-band responses to strengthen frequency components useful for tiny-object representation and reduce indiscriminate mixing between heterogeneous frequency bands.
3) FSSI establishes bidirectional frequency–spatial interaction: frequency-domain responses guide adaptive spatial feature sampling, while spatial context modulates the frequency-domain representation.
4) A frequency-domain structural constraint combines RGB-phase consistency, infrared-confidence regularization, and cross-domain diversity based on absolute cosine similarity to regulate infrared contributions and reduce representation redundancy.

The remainder of this article is organized as follows. Section II reviews related works on multimodal object detection and frequency-domain analysis. Section III elaborates on the proposed FADN framework and its core modules. Section IV presents the experimental results and analyses, and Section V concludes this work.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Single-Modal Remote Sensing Tiny-Object Detection From UAV Perspective

UAV-based tiny-object detection remains challenging because of low spatial resolution, dense target distributions, and substantial scale variations. Akyon et al. [25] introduced a slicing-aided inference framework that processes overlapping image patches to improve small-object recall without modifying the detector. Yang et al. [26] proposed QueryDet, which uses cascaded sparse queries to guide high-resolution feature computation. Wang et al. [27] modeled bounding boxes as 2-D Gaussian distributions and introduced the normalized Wasserstein distance to reduce the sensitivity of tiny-object matching to small positional deviations. Zhu et al. [28] developed TPH-YOLOv5 by combining a transformer prediction head with attention mechanisms to address scale variation and dense occlusion in UAV imagery.

Infrared small-object detection has also motivated architectures for preserving weak target responses. UIU-Net [29] embeds small U-Net structures within a larger U-Net framework to learn multilevel and multiscale features while enhancing global and local contrast. However, UIU-Net performs pixel-level detection on single-frame infrared images, whereas FADN performs bounding-box detection using paired RGB–IR inputs. It is, therefore, discussed as a relevant single-modal method rather than a directly comparable multimodal detector.

Although these methods improve tiny-object representation, they either rely on a single sensing modality or do not explicitly address cross-modal RGB–IR interaction. This motivates the use of complementary infrared information when visible observations are degraded.

B. Visible–Infrared Multimodal Object Detection

Visible–infrared object detection exploits complementary visible texture and infrared thermal information to improve perception under challenging imaging conditions [20], [30]. Tang et al. [31] proposed PIAFusion, which estimates illumination conditions to guide adaptive cross-modal fusion. Huang and Ma [32] introduced a cross-modality fusion transformer that models intramodal and intermodal dependencies within a two-stream backbone. Zhang et al. [33] developed SuperYOLO, combining compact multimodal fusion with an auxiliary superresolution branch to enhance low-resolution remote sensing features.

Most existing RGB–IR detectors perform cross-modal interaction through feature weighting, spatial attention, or transformer-based fusion on spatial feature tensors. As a result, structural cues and modality-dependent energy responses remain implicitly entangled, making it difficult to control unreliable infrared contributions explicitly. On the contrary, FADN decomposes frequency-domain features into amplitude and phase components and assigns asymmetric roles to the two modalities. The visible phase serves as the dominant structural anchor, while infrared amplitude information and a bounded phase residual are incorporated according to infrared confidence.

C. Frequency-Domain Analysis and Modulation in Object Detection

Frequency-domain modeling has been introduced into object detection as a complementary representation to spatial-domain learning. Gu et al. [34] exploited Fourier rotational equivariance to align directional features for oriented object detection. Zhao et al. [35] used frequency-domain modulation and cross-view consistency to improve domain robustness in multicamera 3-D detection. Liu et al. [36] developed unified spatial–frequency modeling for multiscale small-object detection, while Wen et al. [37] introduced frequency-aware attention to suppress background interference in remote sensing tiny-object detection.

More recent methods provide direct References for frequency-aware processing. FreDFT [38] introduces a frequency-domain fusion transformer for visible–infrared object detection, whereas FreqFusion [39] employs adaptive frequency filtering and feature resampling for multiscale feature fusion in dense prediction.

Although these methods demonstrate the value of frequency-domain representations, they generally treat frequency information as a generic enhancement cue and do not explicitly assign modality-specific roles to amplitude and phase or regulate unreliable infrared contributions. FADN instead establishes modality-asymmetric and confidence-controlled amplitude–phase interaction. UACM retains the visible phase as the dominant structural anchor, RFBR recalibrates radial frequency-band responses, and FSSI couples frequency-domain responses with spatial feature learning. Frequency-domain modeling in FADN therefore complements rather than replaces the spatial-domain detector.

D. Spectral–Spatial Representation and Prior-Guided Remote Sensing Learning

Recent remote sensing studies have increasingly emphasized structured representation learning and the integration of prior knowledge with data-driven models. Hong et al. [42] provided a comprehensive overview of hyperspectral imaging and the relationships among sensing characteristics, signal representations, and learning models. SpectralGPT [43] introduced a foundation-model framework for learning transferable spectral–spatial representations from large-scale hyperspectral data. Li et al. [44] further combined disentangled mathematical priors with data-driven representations for hyperspectral anomaly detection, demonstrating the value of embedding structured prior knowledge into trainable networks.

The spectral dimension in hyperspectral imaging represents wavelength-dependent measurements, whereas FADN operates on the 2-D Fourier spatial–frequency representation of RGB–IR feature maps. Although these studies address different signal dimensions and downstream tasks, they share the general principle of combining structured representations and interpretable priors with data-driven learning. Following this perspective, FADN introduces task-specific frequency-domain priors for RGB–IR tiny-object detection through modality-asymmetric interaction, radial frequency-band modeling, and frequency–spatial coupling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

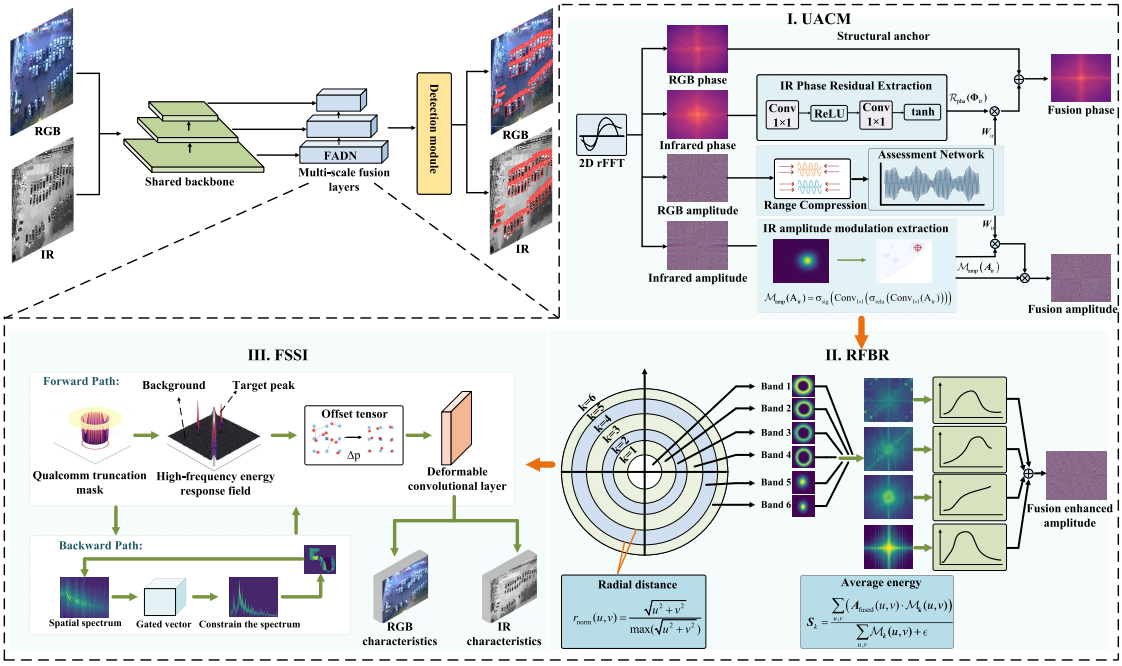


Fig. 2. Architecture of FADN. In UACM, the RGB phase provides the structural anchor, while the infrared phase contributes a bounded, confidence-controlled residual. The infrared-confidence map $\boldsymbol{W}_{\text{ir}}$ regulates both the amplitude-modulation and phase-residual paths.

III. Method

A. Framework

The overall architecture of FADN is illustrated in Fig. 2. FADN introduces complementary frequency-domain interaction at multiple feature scales of the RGB–IR detection framework. At each selected fusion scale, the paired RGB and infrared features are transformed into amplitude and phase components. UACM retains the visible phase as the dominant structural anchor while incorporating confidence-controlled infrared amplitude information and a bounded phase residual. RFBR subsequently recalibrates the responses of different radial frequency bands. After frequency-domain reconstruction and inverse transformation, FSSI couples the resulting representation with spatial features through frequency-guided adaptive sampling and spatial-context feedback. The enhanced multiscale features are delivered to the detection head, and the complete network is jointly optimized by the detection objective and the frequency-domain structural constraint.

B. Uncertainty-Aware Phase–Amplitude Cross-Modulation

Traditional multimodal feature fusion methods generally adopt a global static weight allocation strategy, performing indiscriminate fusion operations on infrared and visible light features at different spatial locations. In UAV remote sensing observation tasks, influenced by dynamic changes in environmental illumination and surface temperature, the infrared radiation contrast between target and background regions significantly decreases, substantially narrowing the difference in their infrared features. If the fusion of infrared and visible light features is still directly executed in the spatial domain under such conditions, it not only fails to leverage infrared features to enhance the target's distinguishability but also introduces background noise from the infrared image into the fusion result, causing interference with the high-resolution structural information carried by the visible light features.

To adaptively evaluate the reliability of infrared information, this article introduces an uncertainty-aware mechanism based on local spectral energy. Rather than indiscriminately mixing the two modalities, the network estimates an infrared confidence map and uses it to regulate the contribution of infrared information during frequency-domain interaction. The visible phase is retained as the dominant structural reference, while reliable infrared information is selectively introduced as complementary amplitude enhancement and a bounded phase residual. The structural diagram is shown as UACM in Fig. 2.

In particular, assuming the preprocessed visible light and infrared remote sensing spatial features are, respectively, represented as $\boldsymbol{F}_{\text{rgb}}$ and $\boldsymbol{F}_{\text{ir}}$, their respective amplitudes and phases are first extracted through a 2-D Fourier transform

$$\boldsymbol{A}_m = |\mathcal{F}(\boldsymbol{F}_m)|, \quad \boldsymbol{\Phi}_m = \angle\mathcal{F}(\boldsymbol{F}_m) \tag{1}$$

where $\mathcal{F}(\cdot)$ denotes the 2-D discrete Fourier transform, and $|\cdot|$ and $\angle(\cdot)$ calculate the modulus and argument of the complex spectrum, respectively.

An uncertainty evaluation network $\mathcal{H}_{\text{conf}}$ is constructed using the logarithmic spectrum amplitude to dynamically predict the confidence weight $\boldsymbol{W}_{\text{ir}}$ of the infrared modality

$$\boldsymbol{W}_{\text{ir}} = \text{Softmax}\left(\mathcal{H}_{\text{conf}}\left(\left[\log\left(1+\boldsymbol{A}_{\text{rgb}}\right), \log\left(1+\boldsymbol{A}_{\text{ir}}\right)\right]\right)\right)_{[:,1,:,:]} \tag{2}$$

where $\log(1+\cdot)$ is utilized to compress the dynamic range of the spectrum amplitude to stabilize network training; $[\cdot,\cdot]$ represents concatenation along the channel dimension; and the subscript $[:,1,:,:]$ denotes extracting the probability map representing the infrared modality from the Softmax output.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

During cross-modulation, the RGB phase is employed as the dominant structural reference, whereas the infrared modality contributes confidence-controlled amplitude information and a bounded phase residual. The nonlinear projection mappings for infrared amplitude modulation and phase residual, $\mathcal{M}_{\text{amp}}(\boldsymbol{A}_{\text{ir}})$ and $\mathcal{R}_{\text{pha}}(\boldsymbol{\Phi}_{\text{ir}})$, are defined as

$$\begin{cases}\mathcal{M}_{\text{amp}}\left(\boldsymbol{A}_{\text{ir}}\right)=\sigma_{\text{sig}}\left(\text{Conv}_{1\times1}\left[\sigma_{\text{ReLU}}\left(\text{Conv}_{1\times1}\left(\boldsymbol{A}_{\text{ir}}\right)\right)\right]\right)\\ \mathcal{R}_{\text{pha}}\left(\boldsymbol{\Phi}_{\text{ir}}\right)=\tanh\left(\text{Conv}_{1\times1}\left[\sigma_{\text{ReLU}}\left(\text{Conv}_{1\times1}\left(\boldsymbol{\Phi}_{\text{ir}}\right)\right)\right]\right)\end{cases} \tag{3}$$

where $\text{Conv}_{1\times1}$ denotes a 1×1 pointwise convolution, and σ_{ReLU} and σ_{sig} denote the ReLU and sigmoid activation functions, respectively. The hyperbolic tangent function constrains the infrared phase residual to $[-1,1]$ radians. The weights and bias of the output projection in the phase-residual branch are initialized to zero, such that the initial phase residual is zero.

The final fused amplitude and phase are calculated as

$$\begin{cases}\boldsymbol{A}_{\text{fused}}=\boldsymbol{A}_{\text{rgb}}\odot\left(1+\boldsymbol{W}_{\text{ir}}\odot\mathcal{M}_{\text{amp}}\left(\boldsymbol{A}_{\text{ir}}\right)\right)\\ \boldsymbol{\Phi}_{\text{fused}}=\text{wrap}\left(\boldsymbol{\Phi}_{\text{rgb}}+\boldsymbol{W}_{\text{ir}}\odot\mathcal{R}_{\text{pha}}\left(\boldsymbol{\Phi}_{\text{ir}}\right)\right)\end{cases} \tag{4}$$

where $\odot$ denotes elementwise multiplication. The circular mapping is defined as

$$\text{wrap}\left(x\right)=\text{atan2}\left(\sin x,\cos x\right) \tag{5}$$

where the function maps each phase value onto the principal interval $(-\pi,\pi]$ while preserving phase periodicity. When $\boldsymbol{W}_{\text{ir}}=0$, the reconstructed amplitude and phase satisfy $\boldsymbol{A}_{\text{fused}}=\boldsymbol{A}_{\text{rgb}}$ and $\boldsymbol{\Phi}_{\text{fused}}=\boldsymbol{\Phi}_{\text{rgb}}$, respectively. Thus, the RGB phase provides the structural basis for phase reconstruction, while the infrared contribution is adaptively regulated by $\boldsymbol{W}_{\text{ir}}$. The fused spectrum is subsequently reconstructed from $\boldsymbol{A}_{\text{fused}}$ and $\boldsymbol{\Phi}_{\text{fused}}$ and transformed back into the spatial domain for subsequent feature processing.

C. Radial Frequency-Band Recalibration

Existing attention mechanisms highly rely on global average pooling or max pooling in the spatial domain. From a UAV remote sensing perspective, tiny objects usually occupy only a few dozen pixels. During the pooling operation, these weak high-frequency features are instantly diluted by low-frequency homogeneous backgrounds, such as water surfaces and grasslands, which occupy an absolute pixel area, ultimately leading to the failure of the attention mechanism.

Given that the spatial scale of the target manifests as the radial distribution of energy in the frequency domain, RFBR provides complementary frequency-domain modeling by explicitly characterizing the responses of different radial frequency bands. Instead, it partitions the spectrum into K concentric bands in polar coordinates, independently models the energy of each frequency band, and learns the relative importance of different bands. This explicitly enhances the high-frequency bands corresponding to tiny objects and suppresses the low-frequency bands of the background. The structural diagram is shown as RFBR in Fig. 2.

In particular, the normalized frequency-domain radial distance $r_{\text{norm}}(u,v)=((u^2+v^2)^{1/2})/(\max((u^2+v^2)^{1/2}))$ is first defined, where (u,v) represents the 2-D coordinate points in the frequency domain. The spectrum is symmetrically divided from the center into K concentric rings, and the binary mask for the kth frequency band is defined as

$$\mathcal{M}_k\left(u,v\right)=\mathbb{I}\left(\lfloor r_{\text{norm}}\left(u,v\right)\cdot K\rfloor=k\right) \tag{6}$$

where $\mathbb{I}(\cdot)$ is an indicator function that equals 1 when the condition is met and 0 otherwise, and $\lfloor\cdot\rfloor$ denotes the floor operation.

In the experiments, the number of frequency bands K is set to 6, as supported by the sensitivity analysis in Section IV-D.

The concentric band division in polar coordinates naturally corresponds to structures of varying scales in remote sensing images: low-frequency bands correspond to large-area backgrounds and the overall contours of targets, while high-frequency bands correspond to detailed features such as target edges and textures. For UAV tiny objects with minimal pixel dimensions, their energy is predominantly concentrated in the high-frequency bands. Even if their spatial proportion is extremely low, their frequency-domain features remain prominent. This provides the frequency-domain motivation for using radial band recalibration as a complementary mechanism to spatial feature learning.

Second, the average energy vector $\boldsymbol{S}_k$ of each frequency band is extracted

$$\boldsymbol{S}_k=\frac{\sum_{u,v}\left(\boldsymbol{A}_{\text{fused}}\left(u,v\right)\cdot\mathcal{M}_k\left(u,v\right)\right)}{\sum_{u,v}\mathcal{M}_k\left(u,v\right)+\epsilon} \tag{7}$$

where ϵ is a minimal constant to prevent division-by-zero overflow.

Third, a learnable frequency-band positional encoding $\boldsymbol{P}$ is introduced to construct a feature sequence $\boldsymbol{S}=[\boldsymbol{S}_1,\ldots,\boldsymbol{S}_K]^{\text{T}}+\boldsymbol{P}$. $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are generated through linear projections to calculate the frequency-band recalibration coefficient $\boldsymbol{A}_{\text{attn}}$

$$\mathbf{A}_{\text{attn}}=\text{Softmax}\left(\frac{\mathbf{Q}\,\mathbf{K}^{\text{T}}}{\sqrt{d_c}}\right)\mathbf{V}\;\mathbf{W}_{\text{out}} \tag{8}$$

where d_c is the channel dimension scaling factor and $\boldsymbol{W}_{\text{out}}$ is the output linear projection weight.

Finally, the enhanced amplitude spectrum is generated by independently scaling and aggregating each frequency band

$$\boldsymbol{A}_{\text{enhanced}}\left(u,v\right)=\boldsymbol{A}_{\text{fused}}\left(u,v\right)+\sum_{k=1}^{K}\boldsymbol{A}_{\text{attn}}^{(k)}\odot\mathcal{M}_k\left(u,v\right) \tag{9}$$

where $\boldsymbol{A}_{\text{attn}}^{(k)}$ is the dynamic weight assigned to the kth frequency band. This mechanism forces the network to learn the inductive bias of the frequency dimension, enabling it to explicitly focus on the high-frequency bands containing the contour features of tiny objects, thereby stripping away the interference of low-frequency homogeneous remote sensing backgrounds from the underlying logic.

D. Frequency–Spatial Synergistic Interaction Mechanism

To cope with the drastic scale and viewpoint variations of remote sensing targets from a UAV perspective, deformable convolutions are frequently employed to enhance the flexibility of the receptive field. However, global deployment across

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

remote sensing images not only incurs exorbitant computational costs but is also prone to inducing deformation artifacts in flat backgrounds.

To this end, this article proposes a frequency–spatial synergistic interaction module, with its structural diagram shown as FSSI in Fig. 2. FSSI establishes bidirectional interaction between frequency-domain responses and spatial feature learning. In the forward path, frequency-domain responses guide adaptive spatial feature sampling according to the current frequency-responsive representation. In the feedback path, spatial contextual information modulates the frequency-domain representation so that frequency enhancement depends on the current spatial context.

Forward Link: Frequency-Domain Guided Spatial Deformation:

First, a high-pass truncation mask $\mathcal{M}_{\mathrm{HP}} = \mathbb{I}(r_{\mathrm{norm}}(u,v) \geq \tau_{\mathrm{hp}})$ is defined, where τ_{hp} is the high-frequency cutoff threshold. In the experiments, τ_{hp} is set to 0.52, meaning frequency points with a normalized radial frequency greater than or equal to 0.52 are considered high frequencies; this value adapts to the radial frequency-band division of $K = 6$, covering the fourth to sixth high-frequency bands, which precisely corresponds to the characteristic frequency range of UAV tiny objects.

The high-frequency spatial energy response field $\boldsymbol{E}_{\mathrm{h}}$ is extracted via the inverse Fourier transform

$$\boldsymbol{E}_{\mathrm{h}} = \frac{1}{C}\sum_{c=1}^{C}\left|\mathcal{F}^{-1}\left(\boldsymbol{A}_{\mathrm{enhanced}} \odot e^{j\boldsymbol{\Phi}_{\mathrm{fused}}} \odot \mathcal{M}_{\mathrm{HP}}\right)\right| \tag{10}$$

where C is the number of feature channels, and $\mathcal{F}^{-1}$ is the 2-D inverse Fourier transform.

Performing min–max normalization yields $\tilde{\boldsymbol{E}}_{\mathrm{h}}$, using $\tilde{\boldsymbol{E}}_{\mathrm{h}}$ as the modulation scalar, it forcibly constrains the offset tensor Δp and the deformation attenuation factor $\mathbf{S}$ of the deformable convolution at spatial feature locations

$$\begin{cases} \mathbf{S}\left(\tilde{\boldsymbol{E}}_{\mathrm{h}}\right) = (1-\beta)\exp\left(-\gamma \cdot \tilde{\boldsymbol{E}}_{\mathrm{h}}\right) + \beta \\ \Delta p = \mathrm{Conv}_{3\times 3}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{3\times 3}\left(\left[\boldsymbol{F}_{\mathrm{spa}}, \tilde{\boldsymbol{E}}_{\mathrm{h}}\right]\right)\right)\right) \odot \mathbf{S}\left(\tilde{\boldsymbol{E}}_{\mathrm{h}}\right) \\ \boldsymbol{F}_{\mathrm{spa}}^{\mathrm{enhanced}}(p_0) = \sum_{\mathbf{p}_n \in \mathcal{R}} w(p_n) \cdot \boldsymbol{F}_{\mathrm{spa}}(p_0 + p_n + \Delta p(p_0)) \end{cases} \tag{11}$$

where β is the deformation lower bound to prevent the total collapse of deformation in background regions; γ is the hyperparameter controlling the steepness of attenuation; p_0 is the central pixel position; and p_n traverses all sampling points within the local neighborhood $\mathcal{R}$.

Feedback Link: Spatial-Domain Constrained Frequency Response:

To prevent local abnormal artifacts caused by blind high-frequency enhancement, the global context of the fused spatial features is extracted to establish a feedback gating vector $\boldsymbol{g}$

$$\begin{cases} \boldsymbol{g} = \sigma_{\mathrm{sig}}\left(\boldsymbol{W}_g \cdot \mathrm{GAP}\left(\boldsymbol{F}_{\mathrm{spa}}\right) + \boldsymbol{b}_g\right) \\ \boldsymbol{F}_{\mathrm{freq}}^{\mathrm{enhanced}} = \boldsymbol{F}_{\mathrm{freq_spatial}} \odot (1 + \boldsymbol{g}) \end{cases} \tag{12}$$

where GAP denotes the global average pooling operation, $\boldsymbol{W}_g$ and $\boldsymbol{b}_g$ are the weights and biases of the fully connected layer; and $\boldsymbol{F}_{\mathrm{freq_spatial}}$ represents the spatial representation of the frequency-domain features after inverse Fourier transform. This link ensures that the global semantic coherence of the spatial domain can act as a penalty term, inversely restricting overactivation in the frequency domain.

The adaptive spatial sampling in FSSI is designed for feature interaction rather than explicit RGB–IR image registration. FSSI does not estimate cross-modal correspondence or geometric transformation parameters and therefore assumes that the paired RGB–IR inputs have undergone basic geometric alignment. It is not designed to compensate for severe cross-sensor displacement, large parallax, or temporal misalignment.

E. Frequency-Domain Structural Constraint Loss

Conventional object detection losses merely supervise the final output, lacking intermediate constraints on the multimodal remote sensing feature fusion process. Therefore, this article designs a frequency-domain structural constraint loss function $\mathcal{L}_{\mathrm{freq}}$

$$\mathcal{L}_{\mathrm{freq}} = \lambda_{\mathrm{phase}}\mathcal{L}_{\mathrm{phase}} + \lambda_{\mathrm{sparse}}\mathcal{L}_{\mathrm{sparse}} + \lambda_{\mathrm{div}}\mathcal{L}_{\mathrm{div}}. \tag{13}$$

The corresponding coefficients are set to $\lambda_{\mathrm{phase}} = 0.50$, $\lambda_{\mathrm{sparse}} = 0.10$, and $\lambda_{\mathrm{div}} = 0.30$.

1) *Phase Structure Preservation Loss*: Given the periodic wrapping characteristic of the phase angle on $[-\pi, \pi]$, directly applying the L1 distance may cause discontinuous errors near the phase boundary. This article, therefore, employs a cosine difference metric

$$\mathcal{L}_{\mathrm{phase}} = \frac{1}{N_{\mathrm{batch}}}\sum_{i=1}^{N_{\mathrm{batch}}}\left(1 - \cos\left(\boldsymbol{\Phi}_{\mathrm{fused}}^{(i)} - \boldsymbol{\Phi}_{\mathrm{rgb}}^{(i)}\right)\right) \tag{14}$$

where N_{batch} represents the total number of samples and pixels participating in the calculation.

2) *Infrared-Confidence Regularization Loss*: To prevent the infrared-confidence map from remaining in an ambiguous intermediate state, a quadratic confidence regularization term is applied

$$\mathcal{L}_{\mathrm{sparse}} = \frac{1}{N_{\mathrm{batch}}}\sum_{i=1}^{N_{\mathrm{batch}}}\boldsymbol{W}_{\mathrm{ir}}^{(i)} \odot \left(1 - \boldsymbol{W}_{\mathrm{ir}}^{(i)}\right). \tag{15}$$

This term constitutes a parabolic penalty on the weights, encouraging the confidence responses to approach clear selections rather than remaining around ambiguous intermediate values.

3) *Cross-Domain Diversity Loss*: To reduce representation redundancy between the spatial and frequency branches, the absolute cosine similarity between their feature representations is minimized

$$\mathcal{L}_{\mathrm{div}} = \frac{1}{B}\sum_{b=1}^{B}\left|\frac{\left\langle \mathrm{vec}\left(\boldsymbol{F}_{\mathrm{spa}}^{(b)}\right), \mathrm{vec}\left(\boldsymbol{F}_{\mathrm{freq}}^{(b)}\right)\right\rangle}{\left\|\mathrm{vec}\left(\boldsymbol{F}_{\mathrm{spa}}^{(b)}\right)\right\|_2 \left\|\mathrm{vec}\left(\boldsymbol{F}_{\mathrm{freq}}^{(b)}\right)\right\|_2}\right| \tag{16}$$

where B denotes the mini-batch size, $\mathrm{vec}(\cdot)$ denotes flattening the feature tensor into a 1-D vector, $\langle\cdot,\cdot\rangle$ denotes the vector inner product, and $\|\cdot\|_2$ denotes the ℓ_2 norm. Because the absolute value is applied to the cosine similarity, $\mathcal{L}_{\mathrm{div}} \in [0, 1]$. Its minimum value is zero when the cosine similarity between the spatial- and frequency-domain representations is zero.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

Minimizing this term suppresses both positive and negative correlations, thereby reducing cross-domain representation redundancy without imposing strict orthogonality.

F. Total Loss

The overall optimization objective of this article is formulated as a joint representation of the detection supervision loss and the feature-level frequency-domain structural constraint loss. During the model training phase, the expression for the network's total loss function $\mathcal{L}_{\text{total}}$ is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_{\text{freq}} \sum_{s \in \mathcal{S}} \mathcal{L}_{\text{freq}}^{(s)} \tag{17}$$

where $\mathcal{S} = \{P3, P4, P5\}$ denotes the set of feature scales where FADN is applied, and the overall frequency-domain loss coefficient is set to $\lambda_{\text{freq}} = 0.10$.

This total loss is composed of two core modules. The first part is the conventional object detection loss $\mathcal{L}_{\text{det}}$, utilized for end-to-end supervision of the network's final prediction results.

It consists of the bounding-box regression loss $\mathcal{L}_{\text{box}}$, classification loss $\mathcal{L}_{\text{cls}}$, and distribution focal loss $\mathcal{L}_{\text{dfl}}$, and is defined as

$$\mathcal{L}_{\text{det}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{dfl}} \tag{18}$$

where λ_{box}, λ_{cls}, and λ_{dfl} denote the weighting coefficients of the bounding-box regression, classification, and distribution focal losses, respectively. They are set to 7.5, 0.5, and 1.5 in all experiments.

The second part is the frequency-domain structural constraint loss $\mathcal{L}_{\text{freq}}^{(s)}$. This term is calculated independently and accumulated for each fusion scale s during the feature extraction stage. It can implicitly learn and preserve the physical structures and frequency-domain fidelity of multimodal remote sensing images at the feature level while ensuring foundational detection accuracy.

The complete objective jointly optimizes the detection objective and frequency-domain feature regularization across the three fusion scales.

IV. Experiment

A. Dataset

Experiments are conducted on DroneVehicle, Visible-Thermal Uncrewed Aerial Vehicle (VTUAV)-det, and VEDAI.

1) *DroneVehicle Dataset:* DroneVehicle [45] is a large-scale paired RGB–IR vehicle-detection dataset collected from UAV platforms. It contains 56 878 images, comprising 28 439 paired visible and infrared observations, with oriented bounding-box annotations for five vehicle categories: car, truck, bus, van, and freight-car. The dataset includes challenging illumination, occlusion, and background conditions. Representative examples are shown in Fig. 3.

2) *VTUAV-det Dataset:* VTUAV-det [46] was reconstructed and reannotated for object detection from the visible–thermal UAV tracking benchmark VTUAV. It contains 16 770 aligned RGB–thermal image pairs and 124 869 bounding-box annotations. The images were captured at different flight altitudes and contain substantial target-scale variations, providing a challenging benchmark for multimodal human-target detection. Representative examples are shown in Fig. 4.

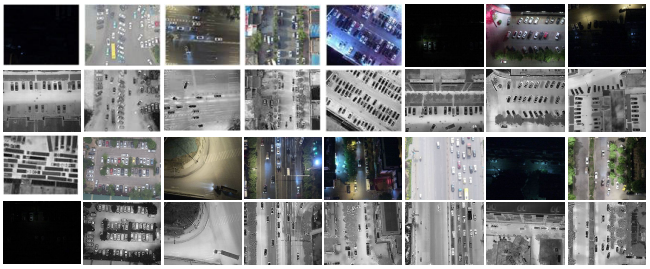


Fig. 3. DroneVehicle dataset.

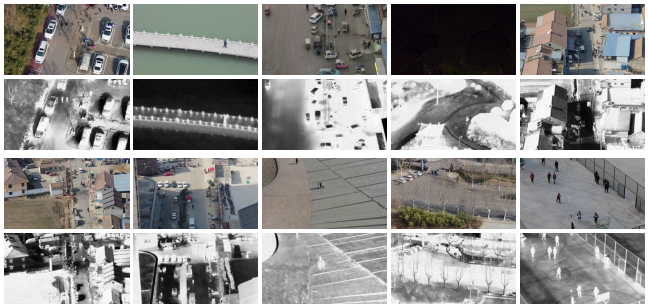


Fig. 4. VTUAV-det dataset.

3) *VEDAI Dataset:* VEDAI [47] provides paired visible and near-infrared aerial images rather than visible and thermal-infrared images. It is, therefore, employed as a complementary RGB–NIR benchmark, while DroneVehicle and VTUAV-det remain the primary RGB–thermal benchmarks.

In our experiments, 1089 paired images are used for training and 121 paired images for validation. The validation set contains 366 annotated instances from nine categories: car, truck, tractor, camping car, van, vehicle, pickup, ship, and plane. All compared models are trained and evaluated using the same data partition to ensure a fair comparison. Since the models are specifically trained on VEDAI, this experiment is regarded as an additional-dataset evaluation rather than a zero-shot cross-dataset transfer experiment.

B. Experimental Setup

FADN and its controlled variants use Ultralytics YOLOv8 as the underlying detector and are trained for 300 epochs using stochastic gradient descent with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 5×10^{-4}. All images are resized to 640×640 pixels. Detection performance is evaluated using mAP@0.5, mAP@0.75, and mAP@0.5:0.95. All experiments are implemented in PyTorch and conducted on a single NVIDIA GeForce RTX 4090-D GPU.

Comparison and Ablation Settings: All comparison methods use the same dataset-specific data partitions, paired multimodal inputs, class definitions, validation sets, and evaluation metrics. FreDFT [38] and FreqFusion [39] are included as frequency-aware comparison methods. Method-specific

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

optimization settings are retained according to their respective implementations. In all controlled ablation experiments, only the evaluated component or configuration is changed, and the prediction head remains unchanged.

Robustness Settings: The spatial-fusion Baseline and FADN are evaluated on the same VTUAV-det validation set using identical annotations and evaluation settings. Corrupted inputs are generated only for evaluation, and neither model is retrained or fine-tuned.

Efficiency Settings: The spatial-fusion Baseline and FADN are evaluated on the same RTX 4090D GPU using 640 × 640 paired inputs, a batch size of 1, and FP32 precision. After 100 warm-up passes, inference latency is averaged over 1000 synchronized forward passes. Training efficiency is measured using an independent controlled 200-epoch benchmark under the same training configuration.

C. Comparative Experiments

To evaluate the proposed FADN, we compare it with representative remote sensing, multimodal, and general object detectors, including RSDet [30], DEYOLO [48], RT-DETRv4 [49], YOLOv11-RGBT [50], and YOLO26 [51].

FreDFT and FreqFusion are additionally included to provide direct comparisons with frequency-aware detection and feature-fusion methods.

1) *Evaluation on the DroneVehicle Dataset:* Table I reports the quantitative results on DroneVehicle. FreDFT achieves an mAP@0.5:0.95 of 63.53%, an mAP@0.5 of 83.93%, and an mAP@0.75 of 75.67%. FreqFusion further improves these metrics to 64.10%, 84.50%, and 76.30%, respectively. FADN achieves the best overall performance, with an mAP@0.5:0.95 of 65.00%, an mAP@0.5 of 84.90%, and an mAP@0.75 of 77.20%. Compared with FreqFusion, FADN improves the three metrics by 0.90, 0.40, and 0.90 percentage points, respectively.

The categorywise results further show that the overall improvement is supported by most object categories. FADN achieves the highest mAP@0.5:0.95 on truck, bus, and van, reaching 63.00%, 80.50%, and 54.50%, respectively, while its 74.30% result on car is tied for the best performance. FADN also achieves the highest mAP@0.5 and mAP@0.75 for car, truck, bus, and van. For the freight-car category, however, RT-DETRv4 remains the strongest method. This result indicates that the proposed method improves the overall multimodal representation without uniformly dominating every individual object category.

Fig. 5 presents representative qualitative comparisons on DroneVehicle. In the selected dense and nighttime scenes, FADN produces fewer missed detections and more complete localization for several distant or partially occluded targets. These examples qualitatively complement the quantitative results in Table I.

2) *Evaluation on the VTUAV-det Dataset:* The quantitative comparison results with state-of-the-art methods on the VTUAV-det dataset are summarized in Table II. On VTUAV-det, FreDFT obtains an mAP@0.5:0.95 of 37.54%, an mAP@0.5 of 63.31%, and an mAP@0.75 of 39.28%. FreqFusion provides a stronger frequency-aware reference, achieving an mAP@0.5:0.95 of 42.30%, an mAP@0.5 of 68.50%, and an mAP@0.75 of 47.00%. FADN further improves the overall performance, achieving an mAP@0.5:0.95 of 45.16%, an mAP@0.5 of 70.23%, and an mAP@0.75 of 49.67%. Compared with FreqFusion, FADN improves the three metrics by 2.86, 1.73, and 2.67 percentage points, respectively.

The categorywise results also show that no method uniformly dominates every category or metric. FADN achieves the highest mAP@0.5:0.95 for the person and uncertain categories and ties YOLO26 for the highest mAP@0.5:0.95 in the crowd category. FADN also achieves the highest mAP@0.5 for the person, rider, and uncertain categories. On the contrary, FreqFusion performs better on the rider category in terms of mAP@0.5:0.95 and mAP@0.75, while YOLO26 retains the highest mAP@0.75 for the crowd category. Therefore, the advantage of FADN is primarily reflected in stronger overall detection performance rather than uniform superiority across every individual category.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET

Method	Category	mAP@.5:.95 (%)	mAP@.5 (%)	mAP@.75 (%)
RSDet	All	54.70	78.70	66.30
	Car	67.00	97.30	82.70
	Truck	47.60	73.60	55.30
	Bus	71.90	94.90	88.00
	Van	42.40	60.70	51.40
	Freight-car	45.40	66.90	54.30
DEYOLO	All	59.40	80.80	70.90
	Car	72.20	97.70	86.60
	Truck	54.30	75.60	63.90
	Bus	76.60	97.00	92.80
	Van	49.60	68.20	60.20
	Freight-car	44.30	65.40	51.20
RT-DETRv4	All	60.90	82.20	72.30
	Car	70.40	97.30	85.50
	Truck	55.20	76.20	63.50
	Bus	76.60	96.80	91.30
	Van	47.60	65.50	56.40
	Freight-car	**54.80**	**75.00**	**64.90**
YOLOv11-RGBT	All	63.10	83.70	75.00
	Car	**74.30**	98.00	88.50
	Truck	59.20	81.60	68.90
	Bus	79.10	97.20	93.90
	Van	53.30	70.90	65.00
	Freight-car	49.60	70.80	58.60
YOLO26	All	59.50	82.80	68.90
	Car	72.50	98.00	86.20
	Truck	56.10	80.40	64.00
	Bus	73.50	95.60	84.90
	Van	49.50	70.30	58.90
	Freight-car	45.80	69.50	50.40
FreDFT	All	63.53	83.93	75.67
	Car	72.58	97.91	87.40
	Truck	61.31	83.23	72.27
	Bus	79.55	97.60	94.57
	Van	53.18	70.69	64.46
	Freight-car	51.04	70.19	59.63
FreqFusion	All	64.10	84.50	76.30
	Car	74.00	98.10	88.60
	Truck	61.70	83.10	73.20
	Bus	80.00	97.60	94.30
	Van	53.20	71.30	64.00
	Freight-car	51.80	72.10	61.30
FADN	All	**65.00**	**84.90**	**77.20**
	Car	**74.30**	**98.20**	**88.90**
	Truck	**63.00**	**84.00**	**74.20**
	Bus	**80.50**	**97.70**	**94.70**
	Van	**54.50**	**72.30**	**65.30**
	Freight-car	52.60	72.30	62.60

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

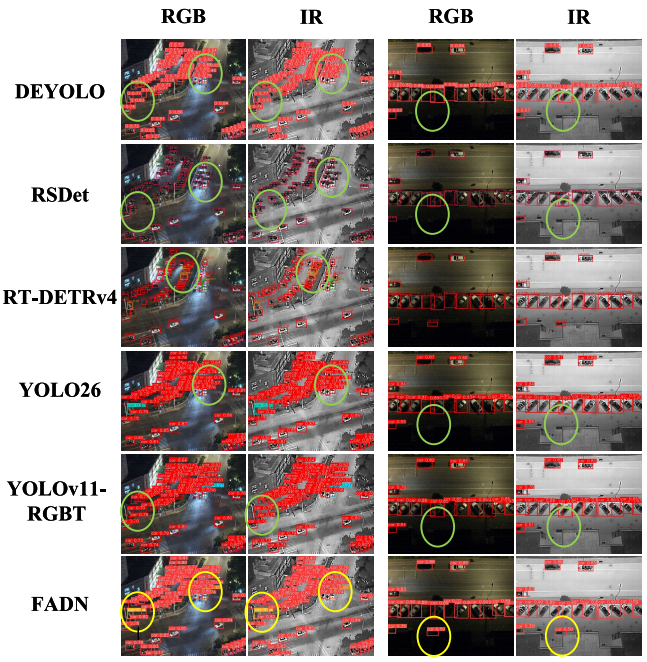


Fig. 5. Visualization of selected methods on the DroneVehicle dataset. Yellow circles indicate the differences between our proposed FADN model and other comparative methods, while green circles mark the regions where the corresponding models exhibit detection deficiencies.

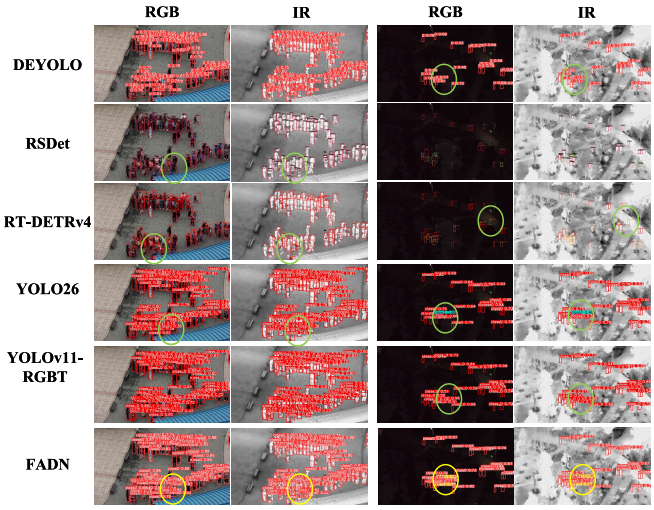


Fig. 6. Visualization of selected methods on the VTUAV-det dataset. Yellow circles indicate the differences between our proposed FADN model and other comparative methods, while green circles mark the regions where the corresponding models exhibit detection deficiencies.

Fig. 6 presents representative qualitative results on VTUAV-det. In the selected dense-crowd scene, FADN detects several highly overlapping instances that are missed by some competing methods. In the low-light example, thermal information provides complementary responses when the visible appearance is degraded. Under UACM, the visible phase remains the structural anchor, while complementary infrared amplitude information and a bounded phase residual are incorporated according to the estimated infrared confidence. These examples qualitatively illustrate the role of frequency-aware multimodal interaction.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VTUAV-DET DATASET

Method	Category	mAP@.5:.95 (%)	mAP@.5 (%)	mAP@.75 (%)
RSDet (2024)	All	36.60	61.00	40.20
	Person	56.30	90.60	62.10
	Rider	41.20	66.90	43.80
	Crowd	45.30	76.50	53.00
	Uncertain	3.80	9.90	1.80
DEYOLO (2025)	All	38.50	64.60	42.40
	Person	63.70	94.30	71.90
	Rider	40.50	67.60	45.20
	Crowd	38.40	73.10	42.70
	Uncertain	11.40	23.30	10.00
RT-DETRv4 (2025)	All	37.80	63.60	38.80
	Person	61.10	93.70	68.30
	Rider	39.80	65.90	37.80
	Crowd	40.40	69.50	45.20
	Uncertain	9.90	25.20	4.10
YOLOv11-RGBT (2025)	All	39.80	65.90	44.40
	Person	62.80	93.90	71.30
	Rider	40.40	70.20	45.10
	Crowd	46.70	**79.40**	52.40
	Uncertain	9.20	20.00	8.61
YOLO26 (2026)	All	40.80	65.60	45.50
	Person	63.20	94.50	72.40
	Rider	40.90	70.40	46.20
	Crowd	**49.50**	76.90	**56.40**
	Uncertain	9.50	20.60	6.98
FreDFT	All	37.54	63.31	39.28
	Person	61.53	94.12	69.16
	Rider	40.51	68.53	41.79
	Crowd	42.11	75.51	42.15
	Uncertain	6.01	15.08	4.00
FreqFusion	All	42.30	68.50	47.00
	Person	66.70	95.30	76.90
	Rider	**49.40**	79.10	**57.40**
	Crowd	42.90	77.40	45.10
	Uncertain	10.20	22.20	8.76
FADN	All	**45.16**	**70.23**	**49.67**
	Person	**67.34**	**95.65**	**78.22**
	Rider	48.49	**80.08**	51.50
	Crowd	**49.50**	79.17	54.40
	Uncertain	**15.30**	**26.03**	**14.56**

TABLE III
OVERALL COMPARISON ON THE VEDAI VALIDATION SET

Method	mAP@.5:.95 (%)	mAP@.5 (%)	mAP@.75 (%)
RSDet	43.50	73.30	51.00
DEYOLO	42.61	68.69	47.72
RT-DETRv4	42.87	70.89	51.43
YOLOv11-RGBT	44.40	72.81	51.43
YOLO26	43.89	69.76	54.09
FreDFT	44.12	66.70	48.90
FreqFusion	44.28	72.62	43.64
FADN	**50.48**	**81.32**	**58.53**

3) *Additional Evaluation on the VEDAI Dataset:* To further evaluate FADN under a different aerial multimodal data distribution and object-category composition, we compare it with RSDet, DEYOLO, RT-DETRv4, YOLOv11-RGBT, YOLO26, FreDFT, and FreqFusion on the VEDAI validation set. For each method, all three aggregate metrics are obtained from the same selected checkpoint on the same validation set.

As shown in Table III, FADN achieves an mAP@0.5:0.95 of 50.48%, an mAP@0.5 of 81.32%, and an mAP@0.75 of 58.53%. Compared with the corresponding second-best results, FADN improves the three metrics by 6.08, 8.02, and 4.44 percentage points, respectively. These results demonstrate that FADN remains effective when independently trained and evaluated on a complementary aerial multimodal dataset with a different object-category composition. Since VEDAI provides

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
OVERALL COMPONENT ABLATION ON THE VTUAV-DET DATASET

Method	UACM	RFBR	FSSI	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
Baseline				65.98	47.93	41.52	74.03	285.50
M1	✓			69.72	49.38	44.18	97.96	359.70
M2		✓		69.91	48.76	43.92	97.68	358.22
M3			✓	67.20	46.65	42.62	103.21	376.10
FADN	✓	✓	✓	**70.23**	**49.67**	**45.16**	104.70	377.61

paired visible and near-infrared images and all compared methods are trained specifically on the same VEDAI data partition, this experiment is regarded as an additional-dataset evaluation rather than validation on a new thermal-infrared dataset or a zero-shot cross-dataset transfer experiment.

D. Ablation Study

To systematically evaluate the contribution of the proposed components, we conduct all ablation experiments on VTUAV-det. The spatial-fusion Baseline performs multimodal feature interaction only in the spatial domain without UACM, RFBR, or FSSI. The prediction head remains unchanged in all configurations.

As shown in Table IV, the spatial-fusion Baseline achieves an mAP@0.5:0.95 of 41.52%. Introducing UACM alone improves the mAP@0.5:0.95 to 44.18%, while RFBR alone reaches 43.92%. FSSI alone provides a smaller but positive improvement, reaching an mAP@0.5:0.95 of 42.62%. When the three components are jointly integrated, FADN achieves an mAP@0.5:0.95 of 45.16%, an mAP@0.5 of 70.23%, and an mAP@0.75 of 49.67%. These results indicate that the three proposed modules provide complementary contributions to the overall detection performance.

1) *Internal Ablation of UACM:* The internal ablation results of the UACM module on the VTUAV-det dataset are presented in Table V. To further investigate the internal mechanism of UACM, we separately evaluate amplitude modulation, bounded phase-residual modeling, and explicit amplitude–phase processing. RFBR and FSSI are disabled in this experiment so that the contribution of UACM can be isolated.

Amplitude-only and phase-only processing improve mAP@0.5:0.95 from 41.52% to 43.65% and 42.71%, respectively, demonstrating that both paths provide useful information. Amplitude modulation produces the larger individual improvement, consistent with its role in incorporating complementary infrared energy information. Direct coupled complex-spectrum fusion achieves 43.18% mAP@0.5:0.95, whereas the complete UACM reaches 44.18%. The absolute improvement of 1.00 percentage point supports explicit amplitude–phase processing under the RGB-phase-anchored formulation.

When UACM is disabled, the neutral frequency-domain reconstruction is constructed from the RGB amplitude and RGB phase, $(\boldsymbol{A}_{\mathrm{rgb}}, \boldsymbol{\Phi}_{\mathrm{rgb}})$, without introducing infrared amplitude or phase information into this frequency-domain reconstruction path.

2) *Effect of Fusion Location:* The detection performance under different fusion location configurations is listed in Table VI. We further investigate the effect of inserting the proposed frequency-domain interaction at different levels of the multiscale feature hierarchy. Single-level configurations at P3, P4, and P5 are compared with the complete P3–P5 configuration.

Among the single-level configurations, P3 achieves the highest mAP@0.5:0.95 of 44.33%, followed by P4 and P5 at 43.82% and 42.95%, respectively. This result suggests that high-resolution shallow features are particularly useful for tiny-object representation. Nevertheless, the complete P3–P5 configuration further improves mAP@0.5:0.95 to 45.16%, indicating that multiscale interaction provides additional benefit by combining fine structural information with deeper semantic representations.

3) *Branch Interaction in FSSI:* The ablation results of the bidirectional interaction mechanism in FSSI are shown in Table VII. To analyze the bidirectional interaction mechanism of FSSI, UACM and RFBR are kept unchanged while the frequency-to-spatial path, the spatial-to-frequency path, or both directions are retained.

The frequency-to-spatial path achieves 44.76% mAP@0.5:0.95, slightly outperforming the spatial-to-frequency path at 44.49%. Enabling both directions further increases mAP@0.5:0.95 to 45.16%, exceeding the two single-direction configurations by 0.40 and 0.67 percentage points, respectively. This result supports the use of bidirectional frequency–spatial interaction in the complete FSSI.

4) *Alternative Frequency Transforms:* The comparison results of different frequency transform approaches are provided in Table VIII. We additionally compare FFT with DCT and DWT as alternative frequency representations. Since DCT and DWT do not provide a complex amplitude–phase representation equivalent to FFT, they are implemented as task-compatible transform-level alternatives rather than as direct replacements of the amplitude and phase variables in UACM.

For each alternative, the corresponding transform-domain coefficients are processed at the same multiscale fusion locations and transformed back to the spatial domain before subsequent detection, while the remaining detector configuration is kept unchanged.

DCT and DWT achieve an mAP@0.5:0.95 of 43.83% and 44.27%, respectively, whereas FFT achieves the highest mAP@0.5:0.95 of 45.16%. FFT is particularly compatible with UACM because it explicitly provides amplitude and phase components that can be processed separately. Although FFT introduces slightly higher computational complexity, it achieves the best detection performance among the evaluated transforms.

5) *Sensitivity to the Number of Radial Frequency Bands:* The sensitivity results of the radial frequency band number K are reported in Table IX. The number of radial frequency bands K is a key hyperparameter of RFBR. We, therefore, evaluate $K \in \{4, 6, 8\}$ while keeping the remaining model settings unchanged.

The results remain stable over the evaluated range. $K = 4$ and $K = 8$ achieve 44.86% and 45.02% mAP@0.5:0.95, respectively, while the default $K = 6$ achieves the highest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

TABLE V
INTERNAL ABLATION OF UACM ON VTUAV-DET

Configuration	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
Spatial-fusion Baseline	65.98	47.93	41.52	74.03	285.50
Amplitude-only	68.75	48.91	43.65	86.15	322.60
Phase-only	67.42	48.35	42.71	85.84	322.35
Coupled complex-spectrum fusion	68.20	48.58	43.18	86.48	323.10
Full UACM	**69.72**	**49.38**	**44.18**	97.96	359.70

TABLE VI
EFFECT OF DIFFERENT FUSION LOCATIONS ON VTUAV-DET

Fusion Location	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
P3 only	69.25	49.10	44.33	78.62	330.80
P4 only	68.80	48.65	43.82	84.85	319.60
P5 only	67.95	47.86	42.95	95.26	305.90
P3–P5	**70.23**	**49.67**	**45.16**	104.70	377.61

TABLE VII
ABLATION OF THE BIDIRECTIONAL INTERACTION IN FSSI

Configuration	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
Frequency→Spatial	69.90	49.15	44.76	102.35	368.90
Spatial→Frequency	69.62	48.82	44.49	102.70	367.80
Bidirectional FSSI	**70.23**	**49.67**	**45.16**	104.70	377.61

TABLE VIII
COMPARISON OF ALTERNATIVE FREQUENCY TRANSFORMS ON VTUAV-DET

Transform	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
DCT	68.95	48.75	43.83	102.40	368.50
DWT	69.40	49.05	44.27	101.80	365.20
FFT	**70.23**	**49.67**	**45.16**	104.70	377.61

TABLE IX
SENSITIVITY TO THE NUMBER OF RADIAL FREQUENCY BANDS ON VTUAV-DET

K	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)	Parameters (M)	GFLOPs (G)
4	69.89	49.34	44.86	104.68	377.56
6	**70.23**	**49.67**	**45.16**	104.70	377.61
8	70.11	49.50	45.02	104.72	377.67

value of 45.16%. Therefore, $K = 6$ is adopted in the main experiments.

E. Loss-Term and Loss-Weight Analysis

To evaluate the influence of the overall frequency-domain loss weight, λ_{freq} is varied over $\{0.05, 0.10, 0.20\}$ while the internal coefficients and all other model and training settings are kept unchanged.

As shown in Table X, $\lambda_{\text{freq}} = 0.10$ achieves the best performance across all three evaluation metrics, with an mAP@0.5 of 70.23%, an mAP@0.75 of 49.67%, and an mAP@0.5:0.95 of 45.16%. Compared with $\lambda_{\text{freq}} = 0.05$, the mAP@0.5:0.95 increases by 0.49 percentage points. When λ_{freq} is further increased from 0.10 to 0.20, the mAP@0.5:0.95 decreases by 0.69 percentage points. Therefore, $\lambda_{\text{freq}} = 0.10$ is adopted for the complete FADN.

TABLE X
SENSITIVITY ANALYSIS OF THE OVERALL FREQUENCY-DOMAIN LOSS WEIGHT ON VTUAV-DET

λ_{freq}	mAP@.5 (%)	mAP@.75 (%)	mAP@.5:.95 (%)
0.05	70.18	48.29	44.67
0.10	**70.23**	**49.67**	**45.16**
0.20	70.13	49.26	44.47

TABLE XI
CONTROLLED ABLATION OF INDIVIDUAL FREQUENCY-DOMAIN STRUCTURAL LOSS TERMS ON VTUAV-DET

Configuration	mAP@0.5	mAP@0.75	mAP@0.5:0.95	ΔmAP@0.5:0.95
All auxiliary terms	70.23	49.67	45.16	–
w/o $\mathcal{L}_{\text{phase}}$	69.23	46.60	44.16	-1.00
w/o $\mathcal{L}_{\text{sparse}}$	70.22	49.66	45.08	-0.08
w/o $\mathcal{L}_{\text{div}}$	69.89	51.49	44.91	-0.25

We additionally conduct a separate controlled ablation to examine the relative contribution of the three auxiliary frequency-domain loss terms. All configurations within this ablation follow the same experimental protocol, and only one auxiliary term is removed at a time.

As shown in Table XI, removing any individual auxiliary term decreases mAP@0.5:0.95 relative to the complete configuration. Removing $\mathcal{L}_{\text{phase}}$ produces the largest reduction of 1.00 percentage point, followed by $\mathcal{L}_{\text{div}}$, whose removal reduces mAP@0.5:0.95 by 0.25 percentage points. Removing $\mathcal{L}_{\text{sparse}}$ results in a smaller reduction of 0.08 percentage points. These results indicate that all three auxiliary constraints contribute positively, with the phase-consistency term providing the most substantial improvement.

F. Computational Efficiency and Accuracy–Efficiency Tradeoff

We further evaluate the training and inference efficiency of the spatial-fusion Baseline and FADN under controlled settings. The inference measurements include the number of parameters, GFLOPs, network-forward latency, throughput, and peak allocated GPU memory, while the training measurements include the average training time per epoch, total training time, and peak training GPU memory.

For a controlled comparison of training efficiency, the total training time reported in this section is measured over a fixed 200-epoch efficiency run for both the spatial-fusion Baseline and FADN, whereas the main detection experiments follow the 300-epoch training protocol described in Section IV-B.

As shown in Table XII, FADN increases the parameter count from 74.03 to 104.70 M and the computational complexity from 285.50 to 377.61 GFLOPs, corresponding to increases of approximately 41.43% and 32.26%, respectively. Its mean network-forward latency increases from 12.079 to 59.517 ms per paired input, approximately 4.93 times that of the Baseline. The corresponding throughput decreases from 82.788 to 16.802 pairs/s, while the peak allocated inference memory increases from 392.435 to 615.618 MiB.

Within the controlled 200-epoch efficiency run, the spatial-fusion Baseline requires 10.05 min per epoch and 33.50 h in total, whereas FADN requires 11.00 min per epoch and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
TRAINING AND INFERENCE EFFICIENCY OF THE SPATIAL-FUSION BASELINE AND FADN

Model	Params. (M)	GFLOPs	Train/Epoch (min)	Total Train (h)	Train Mem. (GB)	Latency (ms/pair)	Throughput (pairs/s)	Infer. Mem. (MiB)
Spatial-fusion Baseline	74.03	285.50	10.05	33.50	9.0	12.079	82.788	392.435
FADN	104.70	377.61	11.00	36.67	20.5	59.517	16.802	615.618

TABLE XIII
CONTROLLED MODALITY-DEGRADATION EVALUATION ON VTUAV-DET

Input Condition	Baseline			FADN			Gain
	mAP@.5	mAP@.75	mAP@.5:.95	mAP@.5	mAP@.75	mAP@.5:.95	(pp)
Clean	65.98	47.93	41.52	70.23	49.67	45.16	+3.64
Visible attenuation, $\beta = 0.50$	65.78	48.14	41.61	69.88	49.09	44.39	+2.78
Visible attenuation, $\beta = 0.25$	63.35	45.15	40.45	68.66	48.08	44.18	+3.73
Visible noise, $\sigma = 15$	65.79	47.04	41.43	70.01	49.51	45.02	+3.59
Visible noise, $\sigma = 30$	65.23	45.87	40.95	69.15	48.93	44.55	+3.60
Infrared noise, $\sigma = 15$	65.37	46.56	41.11	69.45	46.84	44.14	+3.03
Infrared noise, $\sigma = 30$	63.78	40.52	38.64	67.74	44.13	41.56	+2.92
Synthetic fog, $\alpha = 0.20$	64.75	45.92	40.63	68.68	46.70	43.75	+3.12
Synthetic fog, $\alpha = 0.40$	62.19	41.80	38.72	65.44	45.06	41.37	+2.65

TABLE XIV
SENSITIVITY TO CONTROLLED INFRARED-IMAGE TRANSLATION ON VTUAV-DET

Offset	Model	mAP@.5	mAP@.75	mAP@.5:.95	Gain (pp)
0 px	Baseline	65.98	47.93	41.52	–
	FADN	70.23	49.67	45.16	+3.64
4 px	Baseline	63.23	23.33	29.73	–
	FADN	68.16	30.47	34.43	+4.70
8 px	Baseline	48.34	5.66	14.79	–
	FADN	57.01	7.95	17.91	+3.12
16 px	Baseline	13.53	0.37	3.04	–
	FADN	16.76	0.30	3.69	+0.65

TABLE XV
TARGET-SIZE-STRATIFIED EVALUATION ON VTUAV-DET

Size	Num.	Model	mAP@.5	mAP@.75	mAP@.5:.95	Gain (pp)
< 32	495	Baseline	23.03	9.13	10.30	–
		FADN	30.04	10.49	12.59	+2.29
32–64	1773	Baseline	61.05	42.73	37.47	–
		FADN	66.89	44.41	42.79	+5.32
64–96	1587	Baseline	46.41	39.34	32.86	–
		FADN	52.82	43.76	36.23	+3.37
≥ 96	1225	Baseline	55.09	48.66	39.96	–
		FADN	55.40	52.35	41.91	+1.95

36.67 h. The average training time per epoch therefore increases by approximately 9.45%. Peak training GPU memory increases from 9.0 to 20.5 GB.

The additional computational cost is mainly associated with the multiscale Fourier transformation, amplitude–phase modulation, radial frequency-band recalibration, and frequency–spatial interaction. These operations increase both computational complexity and intermediate activation storage, which is particularly reflected in the increased GPU-memory requirement during training.

In terms of the accuracy–efficiency tradeoff, FADN improves mAP@0.5:0.95 from 41.52% to 45.16%, corresponding to an absolute gain of 3.64 percentage points. This improvement is accompanied by a 41.43% increase in parameter count, a 32.26% increase in GFLOPs, and a 4.93-fold increase in forward-pass latency. Therefore, the current FADN should be regarded as an accuracy-oriented multimodal detector rather than a lightweight or edge-ready architecture.

G. Robustness Analysis

1) *Modality-Specific Degradation:* The robustness evaluation results under controlled modality-specific degradation are summarized in Table XIII. We first evaluate the effect of controlled modality-specific degradation. Controlled degradation is independently introduced into the visible or infrared modality while the other modality remains unchanged. This protocol evaluates detector behavior when the quality of one sensing modality deteriorates.

Visible-light attenuation is implemented as

$$I'_{\text{vis}} = \beta I_{\text{vis}}, \quad \beta \in \{0.50, 0.25\}. \tag{19}$$

Gaussian noise is independently added to either modality as

$$I'_m = \text{clip}\left(I_m + n, 0, 255\right), \quad n \sim \mathcal{N}\left(0, \sigma^2\right) \tag{20}$$

where $m \in \{\text{vis}, \text{ir}\}$ and $\sigma \in \{15, 30\}$ in the 8-bit pixel domain. Synthetic visible fog is generated as

$$I_{\text{fog}} = (1 - \alpha)\,\mathcal{G}\left(I_{\text{vis}}\right) + \alpha A \tag{21}$$

where $\mathcal{G}(\cdot)$ denotes Gaussian smoothing, $A = 255$ represents the atmospheric-light intensity in the 8-bit pixel domain, and $\alpha \in \{0.20, 0.40\}$ controls the fog density. This operation provides a controlled combination of image smoothing and atmospheric light but is not regarded as an exact physical simulation of real fog or independently collected adverse-weather observations.

Across the eight corrupted conditions, excluding clean input, FADN achieves an average mAP@0.5:0.95 of 43.62%, compared with 40.44% for the spatial-fusion Baseline. It maintains higher absolute mAP@0.5:0.95 under every evaluated corruption, with margins ranging from 2.65% to 3.73% points.

Relative to their respective clean-input results, the average mAP@0.5:0.95 decreases by 1.08 percentage points for the Baseline and 1.54 percentage points for FADN. Therefore, these experiments demonstrate higher absolute detection accuracy for FADN under the evaluated corruptions rather than a smaller relative performance decline or invariance to modality degradation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

Strong infrared corruption remains challenging. With infrared Gaussian noise at $\sigma = 30$, FADN decreases from 45.16% to 41.56% mAP@0.5:0.95, while retaining a 2.92-percentage-point absolute advantage over the Baseline.

2) *Controlled Cross-Modal Misregistration:* The detection performance under controlled infrared-image translation offsets is presented in Table XIV. To evaluate sensitivity to residual geometric misregistration, only the infrared image is translated while the visible image and ground-truth annotations remain unchanged. Equal horizontal and vertical offsets of 4, 8, and 16 pixels are applied using reflection padding. This protocol isolates the influence of cross-modal displacement without introducing missing border content as an additional factor.

At a four-pixel displacement per axis, FADN achieves 34.43% mAP@0.5:0.95, exceeding the Baseline by 4.70 percentage points. At eight- and 16-pixel translations, FADN retains higher mAP@0.5:0.95, but both models undergo severe performance deterioration. FADN decreases from 45.16% on aligned inputs to 17.91% and 3.69% at eight- and 16-pixel offsets, respectively.

The sharp decline in mAP@0.75 indicates that geometric inconsistency strongly affects precise localization. At the 16-pixel offset, FADN obtains 0.30% mAP@0.75, compared with 0.37% for the Baseline, showing that FADN does not retain an advantage in every metric under severe displacement. The current framework, therefore, assumes approximately registered multimodal inputs and is not claimed to resolve severe cross-modal misregistration.

3) *Target-Size-Stratified Evaluation:* The size-stratified evaluation results on the VTUAV-det dataset are shown in Table XV. To directly evaluate extremely small targets, we divide the clean validation annotations according to the equivalent side length

$$s = \sqrt{wh} \tag{22}$$

where w and h denote the width and height of a ground-truth bounding box at the 640×640 evaluation scale. Four intervals are considered: $s < 32$, $32 \le s < 64$, $64 \le s < 96$, and $s \ge 96$ pixels. Predictions are assigned to the corresponding interval according to their box area, and class-aware matching is independently evaluated over IoU thresholds from 0.50 to 0.95. These results are referred to as size-stratified AP under equivalent-side bins rather than the official COCO AP_S, AP_M, or AP_L metrics.

For the 495 targets with equivalent side lengths below 32 pixels, FADN improves mAP@0.5 from 23.03% to 30.04% and mAP@0.5:0.95 from 10.30% to 12.59%, corresponding to a 2.29-percentage-point improvement in mAP@0.5:0.95. The largest improvement occurs in the 32–64-pixel interval, where mAP@0.5:0.95 increases from 37.47% to 42.79%, corresponding to a gain of 5.32% points.

FADN improves size-stratified mAP@0.5:0.95 across all four intervals. Nevertheless, mAP@0.75 remains low for targets below 32 pixels, indicating that precise localization of extremely small objects remains challenging.

V. Conclusion

In this article, we propose FADN for UAV-based RGB–IR tiny-object detection, using frequency-domain modeling as a complementary representation to spatial-domain feature learning. UACM retains the visible phase as the dominant structural anchor and incorporates complementary infrared information through confidence-controlled amplitude modulation and a bounded phase residual. RFBR recalibrates radial frequency-band responses, while FSSI establishes bidirectional interaction between frequency-domain responses and spatial features. The frequency-domain structural constraint further regulates multimodal representation learning. Experiments on DroneVehicle and VTUAV-det, together with the complementary VEDAI evaluation, component ablations, and controlled degradation analyses, support the effectiveness of FADN. Nevertheless, the current FADN introduces substantial computational and memory overhead and should therefore be regarded as an accuracy-oriented multimodal detector rather than a lightweight or edge-ready architecture. It has not yet been evaluated on resource-constrained edge platforms. The current framework also assumes approximately registered multimodal inputs, relies on a predefined radial frequency partition, and is evaluated primarily under supervised closed-category and dataset-specific settings. Future work will investigate lightweight frequency-domain operators, parameter sharing, structured pruning, knowledge distillation, hardware-aware acceleration, and practical evaluation on resource-constrained UAV and edge-computing platforms. Alignment-aware frequency modeling, adaptive frequency representations, cross-sensor domain adaptation, multimodal foundation-model pretraining, and open-vocabulary tiny-object detection will also be explored.

References

[1] T. Moranduzzo and F. Melgani, "Detecting cars in UAV images with a catalog-based approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6356–6367, Oct. 2014, doi: 10.1109/TGRS.2013.2296351.

[2] Z. Yan et al., "RF-DET: Refocusing on the small-scale objects using aggregated context for accurate power transmitting components detection on UAV oblique imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 692–711, Feb. 2025, doi: 10.1016/j.isprsjprs.2025.01.005.

[3] J. Zhu, Y. Wu, and T. Ma, "Multi-object detection for daily road maintenance inspection with UAV based on improved YOLOv8," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 16548–16560, Nov. 2024, doi: 10.1109/TITS.2024.3437770.

[4] D. Turner, A. Lucieer, and L. Wallace, "Direct georeferencing of ultrahigh-resolution UAV imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2738–2745, May 2014, doi: 10.1109/TGRS.2013.2265295.

[5] H. Ren et al., "Swarm UAV SAR for 3-D imaging: System analysis and sensing matrix design," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5238316, doi: 10.1109/TGRS.2022.3221775.

[6] H. Fang et al., "SCINet: Spatial and contrast interactive super-resolution assisted infrared UAV target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5006722, doi: 10.1109/TGRS.2024.3471786.

[7] T. Zhang et al., "CCSFuse: Collaborative compensation and selective fusion for UAV-based RGB-IR object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5400314, doi: 10.1109/TGRS.2025.3647293.

[8] M. Zhao, W. Li, L. Li, A. Wang, J. Hu, and R. Tao, "Infrared small UAV target detection via isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5004316, doi: 10.1109/TGRS.2023.3321723.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.

[9] S. Ma, Y. Zhang, L. Peng, C. Sun, B. Ding, and Y. Zhu, "OWRT-DETR: A novel real-time transformer network for small-object detection in open-water search and rescue from UAV aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4205313, doi: 10.1109/TGRS.2025.3560928.

[10] G. Gao, Y. Shang, J. Gao, H. Guo, Y. Dong, and C. Li, "YOLD: You only look denseness for tiny object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5607715, doi: 10.1109/TGRS.2026.3660635.

[11] Q. Hu, Y. Zhang, R. Zhang, F. Xu, and W. Yang, "Beyond dehazing: Learning intrinsic hazy robustness for aerial object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 645614, doi: 10.1109/TGRS.2024.3485682.

[12] H. Fan, Y. Qiao, Y. Zhen, T. Zhao, B. Fan, and Q. Wang, "All-day multi-camera multi-target tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2025, pp. 16892–16901, doi: 10.1109/CVPR52734.2025.01574.

[13] R. Wen Liu et al., "Real-time multi-scene visibility enhancement for promoting navigational safety of vessels under complex weather conditions," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 12, pp. 19979–19994, Dec. 2024, doi: 10.1109/TITS.2024.3454597.

[14] B. Tang et al., "R2PLoc: A region-to-point UAV visual geo-localization framework leveraging hierarchical semantic representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5643818, doi: 10.1109/TGRS.2025.3611657.

[15] Q. Li, Y. Zhang, L. Fang, Y. Kang, S. Li, and X. Xiang Zhu, "DREB-Net: Dual-stream restoration embedding blur-feature fusion network for high-mobility UAV object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621218, doi: 10.1109/TGRS.2025.3543270.

[16] H. Fang, C. Wu, X. Wang, F. Zhou, Y. Chang, and L. Yan, "Online infrared UAV target tracking with enhanced context-awareness and pixel-wise attention modulation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5005417, doi: 10.1109/TGRS.2024.3432108.

[17] Z. Shao et al., "TextRSR: Enhanced arbitrary-shaped scene text representation via robust subspace recovery," *IEEE Trans. Multimedia*, vol. 28, pp. 2550–2563, 2026, doi: 10.1109/TMM.2026.3651034.

[18] R. Lu et al., "MTCNet: Multimodal time-sensitive target fast concealment network for UAV aerial imagery via real-time mask generation," *Expert Syst. Appl.*, vol. 333, Jan. 2027, Art. no. 134088, doi: 10.1016/j.eswa.2026.134088.

[19] Q. Xie, Y. Li, and Y. Kang, "SBS: Enhancing parameter-efficiency of neural representations for neural networks via spectral bias suppression," 2025, *arXiv:2509.07373*.

[20] X. Chen et al., "IGIANet: Illumination guided implicit alignment network for infrared-visible UAV detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 40, no. 4, 2026, pp. 356–364, doi: 10.1609/aaai.v40i4.37298.

[21] H. Wang, X. Wu, Z. Huang, and E. P. Xing, "High-frequency component helps explain the generalization of convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 8681–8691, doi: 10.1109/CVPR42600.2020.00871.

[22] J. Lu, "Frequency regularization: Unveiling the spectral inductive bias of deep neural networks," 2025, *arXiv:2512.22192*.

[23] R. Chen, W. Tang, M. Yang, and W. Kang, "FSFNet: Frequency–spatial-domain fusion network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4700913, doi: 10.1109/TGRS.2025.3650074.

[24] Y. Zhou, Y. Feng, S. Huo, and X. Li, "Joint frequency-spatial domain network for remote sensing optical image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627114, doi: 10.1109/TGRS.2022.3196040.

[25] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Bordeaux, France, Oct. 2022, pp. 966–970, doi: 10.1109/ICIP46576.2022.9897990.

[26] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 13658–13667, doi: 10.1109/CVPR52688.2022.01330.

[27] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.

[28] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Montreal, BC, Canada, Oct. 2021, pp. 2778–2788, doi: 10.1109/ICCVW54120.2021.00312.

[29] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023, doi: 10.1109/TIP.2022.3228497.

[30] T. Zhao, M. Yuan, F. Jiang, N. Wang, and X. Wei, "Removal then selection: A coarse-to-fine fusion perspective for RGB-infrared object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 27, no. 2, pp. 2504–2519, Feb. 2026, doi: 10.1109/TITS.2025.3638627.

[31] L. Tang, J. Yuan, H. Zhang, X. Jiang, and J. Ma, "PIAFusion: A progressive infrared and visible image fusion network based on illumination aware," *Inf. Fusion*, vols. 83–84, pp. 79–92, Jul. 2022, doi: 10.1016/j.inffus.2022.03.007.

[32] X. Huang and G. Ma, "Cross-modality object detection based on DETR," *IEEE Access*, vol. 13, pp. 51220–51230, 2025, doi: 10.1109/ACCESS.2025.3551947.

[33] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415, doi: 10.1109/TGRS.2023.3258666.

[34] C. Gu, L. Chen, L. Gu, and Y. Fu, "Fourier angle alignment for oriented object detection in remote sensing," 2026, *arXiv:2602.23790*.

[35] X. Zhao, Q. Gu, X. Wang, C. Zhou, and N. Ye, "Generalizable multi-camera 3D object detection from a single source via Fourier cross-view learning," in *Proc. 42nd Int. Conf. Mach. Learn. (ICML)*, 2025, pp. 77521–77538.

[36] J. Liu, Y. Wang, Y. Cao, C. Guo, P. Shi, and P. Li, "Unified spatial-frequency modeling and alignment for multi-scale small object detection," *Symmetry*, vol. 17, no. 2, p. 242, Feb. 2025, doi: 10.3390/sym17020242.

[37] Z. Wen et al., "FANet: Frequency-aware attention-based tiny-object detection in remote sensing images," *Remote Sens.*, vol. 17, no. 24, p. 4066, Dec. 2025, doi: 10.3390/rs17244066.

[38] W. Wu, X. Zhang, H. Yin, S. Dai, H. Zhang, and Y. Zhang, "FreDFT: Frequency domain fusion transformer for visible-infrared object detection," *IEEE Trans. Multimedia*, early access, Jul. 13, 2026, doi: 10.1109/TMM.2026.3712630.

[39] L. Chen, Y. Fu, L. Gu, C. Yan, T. Harada, and G. Huang, "Frequency-aware feature fusion for dense image prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10763–10780, Dec. 2024, doi: 10.1109/TPAMI.2024.3449959.

[40] L. Shi, C. Li, Z. Zeng, P. Duan, B. Rasti, and A. Plaza, "Masked self-attention fusion network for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Image Process.*, vol. 35, pp. 346–360, 2026, doi: 10.1109/TIP.2025.3648926.

[41] P. Duan, Y. Luo, X. Kang, and S. Li, "LaMamba: Linear attention Mamba for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 527113, doi: 10.1109/TGRS.2025.3613739.

[42] D. Hong et al., "Hyperspectral imaging," *Nature Rev. Methods Primers*, vol. 6, no. 1, Feb. 2026, Art. no. 19, doi: 10.1038/s43586-026-00470-x.

[43] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024, doi: 10.1109/TPAMI.2024.3362475.

[44] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6883–6896, Apr. 2025, doi: 10.1109/TNNLS.2024.3401589.

[45] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022, doi: 10.1109/TCSVT.2022.3168279.

[46] P. Zhang, J. Zhao, D. Wang, H. Lu, and X. Ruan, "Visible-thermal UAV tracking: A large-scale benchmark and new baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 8876–8885, doi: 10.1109/CVPR52688.2022.00868.

[47] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016, doi: 10.1016/j.jvcir.2015.11.002.

[48] Y. Chen et al., "DEYOLO: Dual-feature-enhancement YOLO for cross-modality object detection," in *Proc. Int. Conf. Pattern Recognit. (ICPR)*, 2024, pp. 236–252.

[49] Z. Liao et al., "RT-DETRv4: Painlessly furthering real-time object detection with vision foundation models," 2025, *arXiv:2510.25257*.

[50] D. Wan et al., "YOLOv11-RGBT: Towards a comprehensive single-stage multispectral object detection framework," 2025, *arXiv:2506.14696*.

[51] R. Sapkota, R. H. Cheppally, A. Sharda, and M. Karkee, "YOLO26: Key architectural enhancements and performance benchmarking for real-time object detection," 2025, *arXiv:2509.25164*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.



Xiaomin Liu (Member, IEEE) received the B.S. degree from the School of Information and Electrical Engineering, China University of Mining and Technology, Xuzhou, China, in 2014, and the Ph.D. degree from the School of Information and Control Engineering, China University of Mining and Technology, in 2021.

She is currently an Associate Professor with the School of Information and Control Engineering, China University of Mining and Technology. Her research interests include coal-fired power generation control, reinforcement learning, and object detection.



Haojie Yin received the B.S. degree from the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China, in 2024, where he is currently pursuing the M.S. degree in control science and engineering.

His research interests include image fusion, image recognition, and machine learning. His primary research interests include deep learning and image processing.



Haoyu Wang (Member, IEEE) received the M.S. and Ph.D. degrees from China University of Mining and Technology, Xuzhou, China, in 2017 and 2021, respectively.

He is currently an Associate Professor with the School of Information and Control Engineering, China University of Mining and Technology. His main research interests include hyperspectral image analysis, metalearning, and transfer learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:41:45 UTC from IEEE Xplore. Restrictions apply.